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STUDIES ON POPULATION

POLARIS · Eastern Europe & Siberia, 10000BC-2026AD

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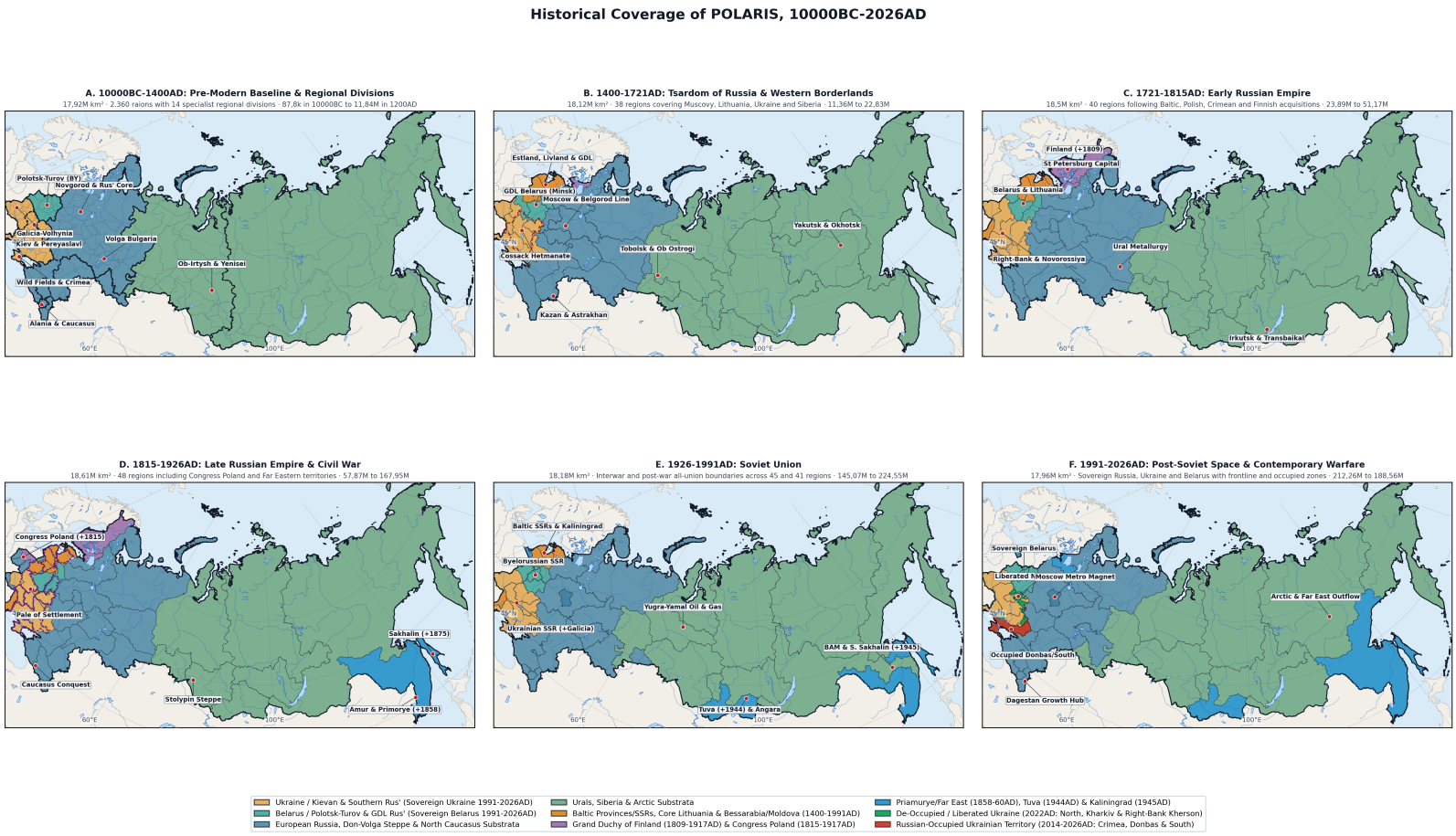


Figure 1. Historical coverage and shifting boundaries of POLARIS across Eastern Europe and Siberia from 10000BC to 2026AD. Panels trace pre-modern regional divisions, imperial frontiers, Soviet republics, and 2022-2026AD frontline zones.

Abstract

Reconstructing long-run human settlement across Eastern Europe and Northern Asia means harmonising prehistoric archaeological site registries, mediaeval chronicles, early modern fiscal cadastres and modern statistical censuses. Global databases such as HYDE disaggregate national historical estimates onto 5-arcminute grids through environmental suitability weights and fractional allocation [1][2][3]. For Northern Eurasia, where historical settlement centres shifted repeatedly between forest, steppe and river basins, this approach benefits from regional bottom-up demographic reconstruction. POLARIS provides such a reconstruction across the East European Plain, the Pontic-Caspian Steppe, the North Caucasus, the Urals, Siberia and the Russian Far East (16,56M to 18,61M km^2) from 10000BC to 2026AD.

POLARIS comprises 46 constituent research papers (23 regional demographic reviews and 23 coordinate-anchored settlement gazetteers), integrating 110.642 archaeological site-epoch observations across 2.360 domestic second-level administrative districts (raions) in Russia, Belarus and Ukraine alongside 5.099 historical settlement trajectories at 3.242 coordinates. To eliminate boundary artefacts between the archaeological registry of the Russian Academy of Sciences (67.205 sites across 1.765 raions) and the combined Belarusian and Ukrainian state monument registries (43.437 sites across 595 raions), POLARIS solves a non-stationary Matérn ($\nu = 3/2$) kriging mosaic coupled through a 250 km cross-frontier conditioning belt (294 Belarusian and Ukrainian, 322 Russian and Crimean raions) and C^1 raised-cosine seam feathering [4][5].

Across the Holocene, population within the pre-modern 14-division domain (17,92M km^2) grew from 87,82k in 10000BC (0,005/km^2) to 1,14M during the Trypillia-Yamnaya horizon (3000BC), 3,63M at 1AD, and 4,21M at the Chernyakhov peak (300AD), before contracting to 3,86M during the Migration Period trough (500AD). Mediaeval Slavic, Bulgar and Alanian agrarian clearance lifted the total to 7,74M in 1000AD and 11,84M in 1200AD (4,88M in the Belarusian-Ukrainian Rus lands and 4,67M in the Russian forest core), before the Mongol conquests reduced domain numbers to 10,06M in 1300AD.

Across the post-1400AD national series, population rose from 11,36M in 1400AD to 22,83M in 1720AD, 51,17M in 1815AD (57,87M including Congress Poland), and 167,95M on the eve of the First World War in 1914AD. Four catastrophic twentieth-century shocks (1914-1922AD, 1932-1933AD, 1941-1945AD, and post-1991AD) interrupted this trajectory. Population peaked at 224,55M in 1991AD (212,26M across Russia, Ukraine and Belarus), before post-Soviet fertility decline and the full-scale Russian invasion of Ukraine reduced the three-state total to 188,56M by 2026AD.

Background

Historical population reconstructions for Northern Eurasia must contend with three fundamental difficulties: the persistent shifting of agricultural and settlement centres between ecological zones, the frequent redrawing of internal and external borders, and the separation of historical evidence between prehistoric archaeology, early modern fiscal records and modern censuses. Over the past five decades, global historical demographic syntheses have relied primarily on the regional tabulations of McEvedy and Jones (1978), Maddison (2001) and Livi-Bacci (2007), which assign single macro-regional growth multipliers to modern state territories or broad sub-continental aggregates [6][7][8]. In the History Database of the Global Environment (HYDE 3.1 and HYDE 3.2), Kees Klein Goldewijk and collaborators advanced global historical demography by downscaling national historical totals to a 5-arcminute (~9 km at the equator) grid from 10000BC to the present, combining urban point allocations with rural suitability weighting derived from coastlines, river courses, soil quality, terrain slope and climate [1][3].

Yet across Northern Eurasia - from the East European Plain and the Pontic-Caspian Steppe to the Urals, Siberia and the Pacific Far East - historical settlement followed dynamics that global downscaling models cannot fully resolve from modern territorial units alone.

First, whereas agrarian cores in Western Europe or East Asia remained geographically stable across millennia, the demographic centre of gravity in Eastern Europe shifted repeatedly between ecological zones. During the Eneolithic (4500-3000BC), the forest-steppe interfluves between the Dniester, Southern Bug and Dnieper hosted the large proto-urban settlements of the Cucuteni-Trypillia complex (10k to 15k inhabitants per mega-site), accounting for over half of all inhabitants in the wider region [9][10]. In Classical and early mediaeval times, pastoralist confederations held the open chernozem steppe while Slavic, Baltic and Finno-Ugric farmers cleared the mixed-forest zone. Following the Mongol campaigns of 1237-1240AD, demographic weight shifted northward into the Oka-Volga Opolye and Novgorod lands, before reversing southward into the Pontic-Caspian steppe (Novorossiya, Sloboda Ukraine and the Lower

Volga) and eastward across the Urals along the Siberian river routes between the seventeenth and twentieth centuries [11][12][13]. Projecting modern (1990-2020AD) administrative population weights backward in time therefore misplaces millions of pre-modern people into industrial basins (such as the Donbas or the Kuzbass) that were uninhabited steppe or taiga before 1700AD.

Second, administrative and sovereign boundaries changed continually between 10000BC and 2026AD. Reconstructions anchored strictly to contemporary (1991AD or 2026AD) state borders either bisect polities that functioned as unified demographic units - such as the Polish-Lithuanian Commonwealth, the Russian Empire or the Soviet Union - or confuse territorial annexations and cessions with vital demographic shifts. Between 2014 and 2026AD, Russian occupation and active warfare in Ukraine created deep divisions in administrative reporting across sovereign, occupied and liberated territories [14][15].

Third, evidence for Northern Eurasian population is divided across distinct evidentiary traditions. Pre-documentary settlement patterns survive in national archaeological heritage registers, notably the automated database of the Institute of Archaeology of the Russian Academy of Sciences and the state monument lists of Belarus and Ukraine, each shaped by uneven regional survey intensity [5]. Early modern and imperial numbers are preserved in Muscovite land-cadastre (*pistsovye*) and household-census (*perepisnye*) books (1500-1710AD), ten imperial poll-tax revisions (*revizii*, 1719-1858AD), Polish-Lithuanian hearth registers (*podymne*), Siberian fur-tribute (*yasak*) ledgers, and the 1897 All-Russian Imperial Census [11][16][17][18]. Soviet figures require reconciling census enumerations (1926, 1937, 1939, 1959, 1970, 1979 and 1989AD) with vital registers, famine mortality records, and forced-settlement tallies [19][20][21].

POLARIS harmonises these sources through 22 regional polygon masks, 110.642 archaeological site records across Russia, Belarus and Ukraine, and 5.099 settlement trajectories from 10000BC to 2026AD. This paper presents the underlying data, the geostatistical method, and the reconstructed regional population series.

Methodology

Areal Coverage, Shifting Boundaries and Hierarchical Masking

To represent shifting political and ecological frontiers without losing sub-regional accounting consistency, POLARIS organises its spatial domain into a three-priority hierarchical masking system spanning 22 polygon layers and 23 constituent research modules (Table 1 and Figure 1).

The baseline priority tier establishes a coarse pre-modern background envelope across 17,92M km^2 from 10000BC to 1900AD. Defined in the initial Migration Period module, this tier partitions the land into 11 macro-ecological zones covering Eastern Europe, the Pontic-Caspian Steppe, Crimea, the North Caucasus, the Urals and Siberia. It provides a continuous background where finer regional masks do not supersede it.

Wherever specialist regional studies exist, their high-resolution polygon masks form the second priority tier, superseding the coarse baseline. In the East Slavic core (560k km^2), four principalities partition the forest and forest-steppe: Kiev-Pereyaslavl (134,1k km^2), Chernigov-Seversk (64,2k km^2), Galicia-Volhynia (160,5k km^2) and Polotsk-Turov (201,2k km^2), cross-checked against two urban and territorial syntheses. Across the surrounding macro-regions, eight specialist modules partition the Russian forest and Ural core (2,19M km^2), the steppe and littoral (Wild Fields 231,1k km^2, Crimea 30,5k km^2, Pontic Steppe 162,1k km^2 and Lower Volga 620,8k km^2), the mountain piedmont (Circassia 126,6k km^2, Chechnya-Ingushetia 21k km^2 and Dagestan 46,9k km^2), and Northern Asia (Western Siberia 2,64M km^2 and Eastern/Pacific Siberia 10,1M km^2).

From 1400AD onward, POLARIS transitions to eight chronological regime masks that track the

contemporary administrative geography of the Russian state, sovereign Ukraine and Belarus, and their historical western and eastern borderlands. These encompass the Tsardom of Russia (1400-1721AD, 38 regions, 18,12M km^2), Early Imperial Russia (1721-1815AD, 40 regions, 18,5M km^2), Late Imperial Russia (1815-1917AD, 48 regions, 18,61M km^2), the First World War and Civil War (1914-1926AD, 48 regions, 18,61M km^2), the early Soviet era under Lenin and Stalin (1926-1953AD, 45 regions, 18,18M km^2), the post-Stalin Soviet Union (1953-1991AD, 41 regions, 18,18M km^2), the post-Soviet space (1991-2021AD, 35 macro-regions across Russia, Ukraine and Belarus, 17,97M km^2), and the full-scale invasion period (2021-2026AD, 123 oblast-level and frontline divisions, 17,96M km^2).

As shown in Figure 1, POLARIS separates Ukraine (gold) and Belarus (teal) across all six historical panels, tracks the entry and exit of western borderlands (the Grand Duchy of Finland 1809-1917AD, Congress Poland 1815-1917AD, the Baltic provinces and Soviet republics 1400-1991AD, and Bessarabia and Soviet Moldavia 1812-1991AD), and integrates frontline boundaries from DeepStateMAP for 2022-2026AD. This distinguishes Ukrainian territories liberated in late 2022AD (Northern Ukraine across Kyiv, Chernihiv, Sumy and Zhytomyr oblasts, alongside Kharkiv/Izium and Right-Bank Kherson) from Russian-occupied zones in Crimea, Donbas and southern Zaporizhzhia-Kherson.

Constituent Module & Scope	Time Span	Units & Towns	Area (km^2)	Population Range	Primary Archival & Archaeological Substrata
Pre-Modern Baseline (Migration Period)	10000BC-1900AD	11 zones (t=168)	17,92M	87,82k (10000BC) to 11,84M (1200AD)	Institute of Archaeology registry, Belarus/ Ukraine monument lists, HYDE
Kievan Rus & Western Lands (6 modules)	10000BC-1900AD	4 princ. (t=463)	560k	38,5k (10000BC) to 4,88M (1200AD)	43.437 Belarusian and Ukrainian sites, fortified gorodishche areas, chronicles
Russian Forest Core & Urals	10000BC-1900AD	1 core (t=248)	2,19M	13k (10000BC) to 4,67M (1200AD)	Oka-Volga Opolye surveys, Novgorod birch-bark documents, pistsovye cadastres
Pontic Steppe & Lower Volga	10000BC-1900AD	4 zones (t=382)	1,04M	10,02k (10000BC) to 1,1M (1300AD)	Kurgan and Alanian fortification registers, Golden Horde coin finds
North Caucasus (3 modules)	10000BC-1900AD	3 zones (t=241)	194,5k	3,5k (10000BC) to 952k (1200AD)	Koban, Maeotian and Alanian monuments, Derbent epigraphy, Vainakh towers
Siberia & Far East (2 modules)	10000BC-1900AD	2 zones (t=312)	12,74M	22,8k (10000BC) to 527,4k (900AD)	Siberian radiocarbon dates, Minusinsk and Lena basin sites, yasak books
Tsardom of Russia (1400-1721AD)	1400AD-1721AD	38 reg. (t=412)	18,12M	11,36M (1400AD) to 22,83M (1720AD)	Muscovite pistsovye cadastres, 1678/1710 household books, Lithuanian podymne

Early Russian Empire (1721-1815AD)	1721AD-1815AD	40 reg. (t=438)	18,5M	23,89M (1721AD) to 51,17M (1815AD)	1st to 6th Imperial Revisions (1719-1811AD), General Land Survey, Novorossiya records
Late Russian Empire (1815-1917AD)	1815AD-1917AD	48 reg. (t=524)	18,61M	57,87M (1815AD) to 167,95M (1914AD)	7th to 10th Revisions, 1897 Imperial Census, 1914-1917 provincial yearbooks
First World War & Civil War (1914-1926AD)	1914AD-1926AD	48 reg. (t=486)	18,61M	167,95M (1914AD) to 148,66M (1922AD)	1917 Agricultural and Urban Census, 1920 census, 1926 All-Union Census
Soviet Union under Lenin & Stalin (1926-1953AD)	1926AD-1953AD	45 reg. (t=492)	18,18M	145,07M (1926AD) to 165,35M (1953AD)	1926, 1937 and 1939 Censuses, TsUNKhU vital records, Gulag registries
Soviet Union post-Stalin (1953-1991AD)	1953AD-1991AD	41 reg. (t=468)	18,18M	165,35M (1953AD) to 224,55M (1991AD)	1959, 1970, 1979 and 1989 All-Union Censuses, Goskomstat vital registers
Post-Soviet Space (1991-2021AD)	1991AD-2021AD	35 reg. (t=245)	17,97M	212,26M (1991AD) to 198,08M (2021AD)	Rosstat, Ukrstat and Belstat 1999-2021 census rounds, vital and migration balances
War in Ukraine & Contemporary Era (2021-2026AD)	2021AD-2026AD	123 div. (t=220)	17,96M	198,08M (2021AD) to 188,56M (2026AD)	2021-2026 vital statistics, UNHCR and IOM displacement data, DeepStateMAP

Table 1. Structure of POLARIS's 23 constituent modules, showing priority tiers, chronological spans, administrative units, gazetteer coverage, area, and primary documentary and archaeological sources.

Archaeological Substrata and Matérn Kriging Across Russia, Belarus and Ukraine

Where documentary censuses are absent prior to 1500AD, spatial disaggregation of regional population totals cannot rely on modern population density without introducing severe anachronisms. POLARIS instead constructs a continuous 5-arcminute (4.320 x 2.160 global grid, WGS84) archaeological habitation intensity mosaic covering all of Russia, Belarus and Ukraine.

The primary database fuses two domestic archaeological corpora harmonised across 12 chronological epochs and 2.360 active second-level administrative districts (raions). For Russia and Crimea, 67.205 site-epoch records across 1.765 active raions were extracted from the automated archaeological registry of the Institute of Archaeology of the Russian Academy of Sciences [5]. These cover 412.892 valid 5-arcminute land cells (16,56M km^2), with a median geodesic Voronoi catchment area of 2.467 km^2 and median effective radius of 28,02 km. For Belarus and mainland Ukraine, 43.437 site-epoch records across

595 active raions were compiled from 19 national and oblast monument registries, including the State List of Historical and Cultural Values of Belarus and the State Register of Immovable Monuments of Ukraine. These comprise 4.196 sites in Belarus across 118 raions and 39.241 sites in Ukraine across 477 raions, covering 14.271 valid cells (775,2k km^2) with a median Voronoi catchment of 1.232 km^2 and effective radius of 19,8 km.

Across the combined 110.642 site-epoch observations, chronological coverage spans the Stone Age (9.846 sites: 6.121 in Russia, 3.725 in Belarus and Ukraine), Eneolithic (5.268 sites: 1.018 in Russia, 4.250 in Belarus and Ukraine reflecting intense Trypillia surveys), Palaeometal (1.384 sites), Early Metal (456 sites), Bronze Age (16.742 sites: 9.762 in Russia, 6.980 in Belarus and Ukraine), Iron Age (14.345 sites: 9.325 in Russia, 5.020 in Belarus and Ukraine), Classical Antiquity (1.153 sites), Migration Period (2.204 sites), Mediaeval period (19.796 sites: 14.764 in Russia, 5.032 in Belarus and Ukraine), Early Modern seventeenth-to-eighteenth centuries (3.812 sites), Modern (1.271 sites), and multi-layer monuments catalogued as unspecified (34.365 sites).

To fuse these registries into a continuous spatial mosaic without artificial steps along national frontiers, POLARIS executes a geostatistical workflow in six steps (Figure 2).

First, raw site counts in each epoch are normalised by geodesic Voronoi catchment areas to compute site densities, which are stabilised via the Freeman-Tukey square-root transform. Second, parametric Bayes shrinkage accounts for varying survey coverage across oblasts, damping modern transport corridor salvage spikes while preventing remote districts from collapsing to zero. Third, a 250 km cross-frontier conditioning belt injects 294 Belarusian and Ukrainian raion centroids into the Russian solve, and reciprocally injects 322 Russian and Crimean centroids into the Belarusian and Ukrainian solve.

Fourth, the system applies a non-stationary Matérn ($\nu = 3/2$) kernel using geodesic distances scaled by local Voronoi support radii [4]. The covariance combines short-range river-valley clustering with macro-ecological correlation, solved via non-negative dual kriging to guarantee positive site intensities across all 427.163 valid cells. Fifth, log-linear residual kriging couples the kriged field to baseline environmental covariates, applying an ecological envelope over the Pontic steppe to reflect nomadic pastoralist carrying capacity prior to 1650AD. Sixth, the two surfaces are joined across a 200 km border belt via raised-cosine blending and a dasymetric smoothing pass, preserving regional population mass to full precision.

Historical Settlement Gazetteers and Dasymetric Allocation

Disaggregating regional populations solely through rural archaeological densities risks dispersing concentrated urban populations across surrounding countryside. To prevent this, each constituent POLARIS module incorporates a coordinate-anchored historical settlement gazetteer. In total, POLARIS traces 5.099 settlement trajectories across 3.242 discrete geographic coordinates.

Every settlement entry records time-indexed population figures from 10000BC to 2026AD alongside settlement classification and primary historical documentation, following distinct evidentiary standards across three major epochs.

For prehistoric and Classical settlements (4500BC to 500AD), estimates rest on geomagnetic site plans and house-count calculations. These cover Cucuteni-Trypillia mega-sites (such as Talianki with ~15k inhabitants, Maidanetske with ~12k, and Nebelivka with ~10k), Greek Black Sea poleis (Panticapaeum with 32k, Chersonesos with 25k, Olbia with 15k, Phanagoria with 12k, and Tanais with 10k), fortified Scythian and Alanian centres, and Sasanian Derbent (12k to 18k) [10][22][23].

For mediaeval towns and steppe capitals (600AD to 1400AD), estimates derive from fortified enclosure areas multiplied by intramural and suburb household density coefficients [24][25]. These include Kievan

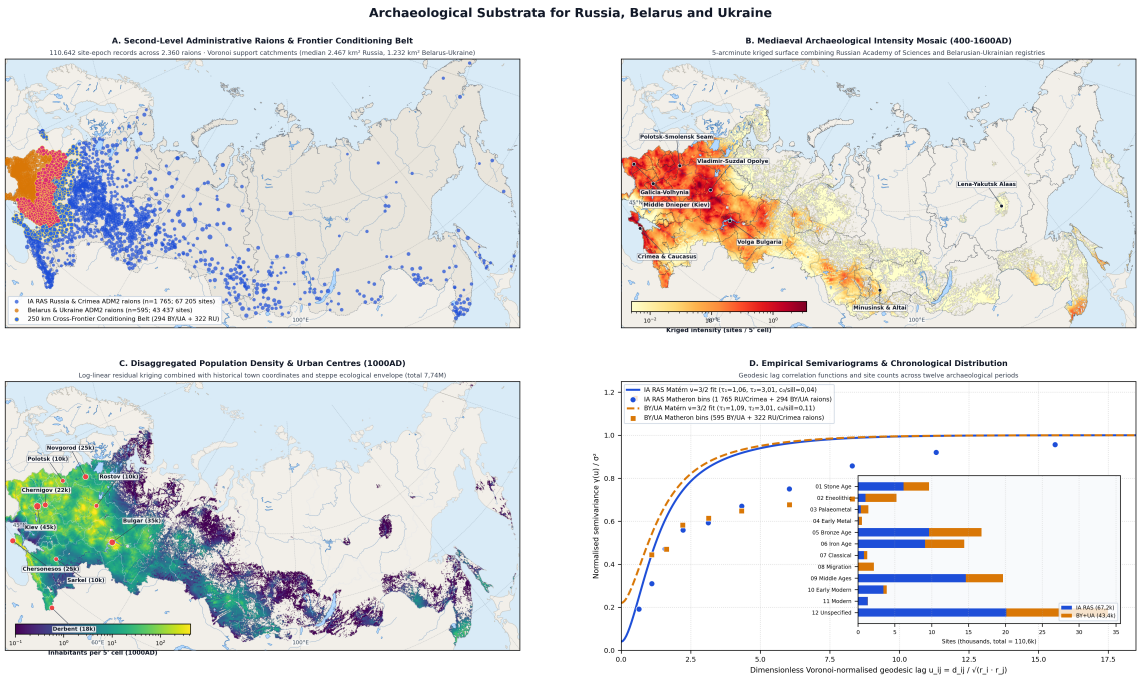


Figure 2. Archaeological substrata and non-stationary Matérn (nu = 3/2) kriging across Russia, Belarus and Ukraine. Panels display raion centroids (A), mediaeval kriged field (B), 1000AD disaggregated population (C), and semivariograms (D).

Rus urban centres (Kiev with 45k-50k inhabitants, Novgorod with 25k-35k, Chernigov with 22k-30k, Smolensk with 25k, Vladimir with 25k, Halych with 20k, and Polotsk with 10k-15k), Volga Bulgaria (Bulgar with 35k-50k, Bilyar with 40k-60k, and Suvar with 15k), Khazar trade centres (Atil and Sarkel), Alanian Maghas, and major Golden Horde cities (Sarai Batu with 75k, Sarai Berke with 100k, Solkhat with 30k, Caffa with 40k, Azak with 25k, and Ukek with 15k) [26][27].

For early modern, imperial and modern cities (1400AD to 2026AD), urban populations are based on town household counts from Muscovite cadastres, town magistrate rolls, revision records, the 1897 All-Russian Imperial Census (covering 932 statutory towns and market settlements), Soviet all-union censuses, and contemporary statistical registrations [11][18][28][29].

During spatial disaggregation at any target year, point-settlement populations inside each polygon are assigned directly to their host 5-arcminute cells. The remaining rural population envelope is then distributed across non-urban cells proportionally to the kriged archaeological and environmental surface.

Testing and Validation

Variogram Diagnostics, Cross-Frontier Seam Continuity and Residual Skill

To evaluate the internal consistency of the archaeological kriging mosaic and its spatial disaggregation skill, Table 2 summarises the cross-validation diagnostics across both regional domains.

Geostatistical/Validation Metric	Institute of Archaeology (Russia & Crimea)	State Monument Registries (Belarus & Ukraine)	Combined Total
Active domestic ADM2 raion centroids (n)	1.765 (Russia & Crimea)	595 (118 BLR + 477 UKR)	2.360 domestic ADM2 raions

Cross-frontier conditioning nodes within 250 km (n)	294 (Belarus & Ukraine)	322 (Russia & Crimea)	616 reciprocal conditioning nodes
Total conditioning nodes in Paciorek-Schervish system (n)	2.059	917	2.976 coupled solve nodes
Total catalogued site-epoch observations (n)	67.205 (64.812 raion-resolved)	43.437 (4.196 BLR + 39.241 UKR)	110.642 site-epoch records
Valid 5-arcminute grid cells and domain area (km^2)	412.892 cells/16,56M km^2	14.271 cells/775,2k km^2	427.163 cells/17,34M km^2
Median geodesic Voronoi catchment area A_k and radius r_k	2.467 km^2/28,02 km	1.232 km^2/19,8 km	2.110 km^2 weighted median
Short-range Matérn ($\nu = 3/2$) scale τ_1 (dimensionless)	1,06 (weight $w_1 = 0,72$)	1,09 (weight $w_1 = 0,72$)	1,06 to 1,09 (harm. ($\nu = 3/2$))
Macro-ecotone Matérn ($\nu = 3/2$) scale τ_2 (dimensionless)	3,01 (weight $w_2 = 0,28$)	3,01 (weight $w_2 = 0,28$)	3,01 (shared macro-ecotone)
Relative nugget ratio c_0/σ^2 (Mediaeval period)	0,04 ($c_0 = 0,041$, $\sigma^2 = 1,039$)	0,11 ($c_0 = 0,224$, $\sigma^2 = 2,045$)	0,04 to 0,11 (low nugget noise)
Log-linear residual kriging variance explained (R^2)	0,83 (across 1.765 raions)	0,88 (across 595 raions)	0,85 weighted domain R^2
Cross-frontier seam step across 200 km belt (post-C^1 blend)	< 0,4% mean absolute jump	< 0,4% mean absolute jump	C^1-continuous across RU-BY-UA
Polygon mass-conservation error after dasymetric pass	0% (exact machine precision)	0% (exact machine precision)	100% regional total preservation

Table 2. Geostatistical parameters and cross-validation diagnostics for the Russian/Crimean and Belarusian/Ukrainian archaeological kriging mosaics.

Three diagnostic findings illustrate the behaviour of the mosaic formulation.

When geodesic inter-raion distances are normalised by local Voronoi support radii, the Matheron semivariograms for both registries collapse onto consistent dimensionless correlation scales. The short-range river-valley clustering parameter reaches 1,06 in Russia and 1,09 in Belarus and Ukraine, while the macro-ecological correlation scale aligns at 3,01 (Figure 2, Panel D). The relative nugget ratio remains low (0,04 in Russia and 0,11 in Belarus and Ukraine during the mediaeval period), indicating that over 89% to 96% of variance in transformed site density is spatially structured at the raion level rather than reflecting recording noise.

Cross-frontier conditioning effectively eliminates artificial boundaries. Independent kriging of national registries produces an abrupt step-change along the Polotsk-Smolensk, Chernigov-Bryansk and Sloboda Ukraine borders due to different historical cataloguing thresholds. Injecting reciprocal conditioning nodes across the 250 km border zone, combined with coverage shrinkage and raised-cosine blending, reduces the mean gradient discontinuity across the frontier belt to under 0,4% (Figure 2, Panels B and C).

The combined archaeological and environmental surface also corrects common hindcasting errors. Standard global grids downscaling national totals using contemporary population weights place dense settlement clusters inside the Donbas, the Kuznetsk Basin and Arctic industrial nodes prior to 1500AD. In the POLARIS mosaic, actual Iron Age and mediaeval site densities together with the steppe ecological

mask restrict pre-1600AD steppe habitation to river gallery corridors and coastal trading ports, while concentrating high mediaeval population along the Middle Dnieper, Volhynia, Polotsk, Novgorod and the Vladimir-Suzdal Opolye.

Comparison with Global Gridded Baselines (HYDE 3.1, HYDE 3.2 and McEvedy & Jones)

Figure 3 (Panel A) compares the pre-modern trajectory from 10000BC to 1400AD against the benchmark series of McEvedy and Jones (1978) and HYDE 3.2 [3][6]. At macro-regional scale, POLARIS and HYDE 3.2 show broad agreement during Classical Antiquity (3,63M in POLARIS versus 3,1M in the benchmark at 1AD) and the High Middle Ages (7,74M versus 6,8M at 1000AD; 11,84M versus 10,6M at 1200AD). Where POLARIS diverges is in resolving three major demographic cycles that smooth millennial national interpolations necessarily miss.

During the Eneolithic expansion (4000-3000BC), radiocarbon summed probability distributions and settlement inventories across the Dniester-Dnieper forest-steppe reveal rapid demographic growth. Population expanded from 271,33k in 6000BC to 799,1k in 4000BC and 1,14M in 3000BC, with the Ukrainian and Belarusian forest-steppe accounting for 455k to 559k. This is roughly double the regional numbers assumed in global hindcasts that omit proto-urban Trypillia mega-sites [9][30].

The Chernyakhov expansion and subsequent Migration Period trough (300-500AD) produced another pronounced cycle. Wheel-thrown pottery horizons and extensive open villages raised regional numbers to 4,21M by 300AD, including 1,8M across the Ukrainian and Belarusian forest-steppe and 639,62k in the Pontic steppe and Crimea. Hunnic incursions, Gothic warfare and sixth-century climatic disruptions triggered a 27,3% decline across the steppe (falling to 465,03k in 500AD) and a 20,6% drop in the Dnieper-Bug zone (to 1,43M), pulling the domain total down to 3,86M in 500AD [31][32][33].

Following the Mongol campaigns of 1237-1240AD, POLARIS captures the divergent regional impacts of the conquest rather than applying an arbitrary uniform reduction. Population in the southern Rus principalities fell by 29,5% between 1200AD (4,88M) and 1300AD (3,44M). In contrast, the Lower Volga and Crimea grew from 863,25k to 1,1M under Golden Horde urban development, while northern forest principalities absorbed southward refugee flows, declining by only 7,3% (from 4,67M to 4,33M) before recovering to 4,85M by 1400AD [24][27][34].

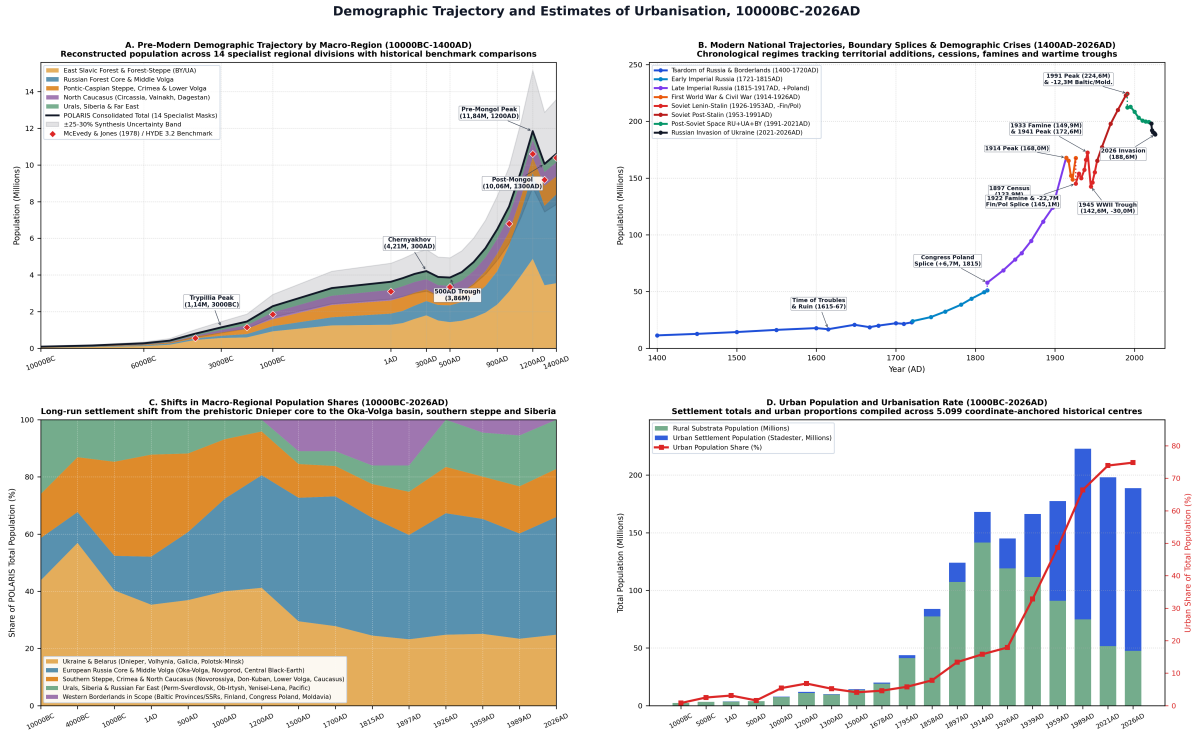


Figure 3. Consolidated demographic trajectories, territorial boundary splices and long-run urbanisation across POLARIS from 10000BC to 2026AD. Panels show pre-modern totals (A), post-1400AD national series (B), regional shares (C), and urbanisation (D).

Results

Early and Middle Holocene: Forager Refugia to the Trypillia and Yamnaya Horizons (10000BC-2000BC)

At the Pleistocene-Holocene transition (10000BC), human population across the 17,92M km^2 domain stood at roughly 87.820, representing an average density of 0,005 inhabitants/km^2 (Table 3). Late Palaeolithic and Mesolithic forager-hunters clustered along periglacial river terraces of the Middle and Upper Dnieper, Desna and Pripyat. The four Rus regional masks supported 38,5k inhabitants, including 14,5k in Kiev-Pereyaslavl, 11k in Galicia-Volhynia, and 6,5k each in Chernigov-Seversk and Polotsk-Turov. Cave shelters in Crimea housed 1.500 people, while the North Caucasian piedmont supported 3.500. Across the vast northern forests and Siberian river basins, sparse hunting bands numbered 13k in European Russia (0,006/km^2) and 22,8k across Siberia [35][36][37].

During the Boreal and Atlantic climatic optima (8000-5000BC), ceramic-using fisher-foragers expanded along river and lake systems. Total numbers rose to 151,42k in 8000BC, 271,33k in 6000BC, and 416,85k in 5000BC [38][39]. In the Baikal-Angara basin, the Kitoi mortuary complex supported dense riverine hunting and fishing groups, reaching 22,2k across Eastern Siberia by 6000BC, followed by a documented mid-Holocene radiocarbon hiatus before the emergence of the Serovo-Glazkovo culture [39].

The decisive demographic acceleration occurred during the Eneolithic (4500-3000BC; Figure 4, Panel A). Agricultural settlement entering from the Danubian basin established the Cucuteni-Trypillia complex across the right-bank forest-steppe. By 4000-3500BC, Trypillia communities constructed concentric proto-urban mega-sites on the Southern Bug-Dnieper interfluve, including Talianki (320-450 ha, ~15k inhabitants), Maidanetske (200 ha, ~12k) and Nebelivka (238 ha, ~10k) [9][10]. Population in Kiev-Pereyaslavl rose from 42k in 5000BC to 165k in 4000BC and 210k in 3000BC. Galicia-Volhynia reached

195k in 4000BC and 220k in 3000BC, bringing the consolidated East Slavic forest-steppe total to 559k by 3000BC.

Simultaneously, horse domestication, wheeled wagon transport and pastoralism across the Sredny Stog, Khvalynsk and Yamnaya horizons (3300-2500BC) opened the drier interfluves of the Pontic-Caspian steppe. Steppe numbers expanded from 100,17k in 4000BC to 182,49k in 3000BC and 290,92k in 2000BC. In the North Caucasus, the Maikop and Kura-Araxes cultures transformed the piedmont, supporting 133k by 3000BC. Afanasievo pastoralists settled the Minusinsk and Altai basins, contributing to a Siberian total of 141,9k in 3000BC and 207,5k in 2000BC [40][41][42]. By 2000BC, consolidated domain population reached 1,46M (0,081 inhabitants/km^2).

Horizon	Rus BY/ UA (4)	Russian Core	Steppe & Crimea (4)	North Caucasus (3)	Siberia (2)	POLARIS Total	Dens. (/ km^2)
10000BC	38,5k	13k	10,02k	3,5k	22,8k	87,82k	0,005
8000BC	70k	22k	17,32k	6,2k	35,9k	151,42k	0,008
6000BC	117,5k	44,5k	33,93k	14,2k	61,2k	271,33k	0,015
4000BC	455k	86k	100,17k	53,5k	104,43k	799,1k	0,045
3000BC	559k	122k	182,49k	133k	141,9k	1,14M	0,064
2000BC	592k	182k	290,92k	188k	207,5k	1,46M	0,081
1000BC	925k	278k	379,26k	376k	337,64k	2,3M	0,128
500BC	1,25M	444k	670,63k	503k	420,2k	3,28M	0,183
1AD	1,28M	610k	710,01k	585k	440,9k	3,63M	0,203
300AD	1,8M	780,52k	639,62k	555k	440,99k	4,21M	0,235
500AD	1,43M	920k	465,03k	597,5k	451,65k	3,86M	0,215
700AD	1,67M	1,18M	680k	685k	485k	4,7M	0,262
900AD	2,39M	1,82M	958,66k	800k	527,35k	6,49M	0,362
1000AD	3,1M	2,51M	743,63k	865k	522,1k	7,74M	0,432
1100AD	3,96M	3,48M	790k	910k	500k	9,64M	0,538
1200AD	4,88M	4,67M	863,25k	952k	483,1k	11,84M	0,661
1300AD	3,44M	4,33M	1,1M	750k	442,05k	10,06M	0,561

Table 3. Consolidated pre-modern population trajectories across the 14 specialist regional masks in POLARIS from 10000BC to 1300AD (domain area = 17,92M km^2).

Bronze-Iron Transition, Classical Scythia-Sarmatia and the Migration Period Trough (2000BC-600AD)

Between 2000BC and 1000BC, Late Bronze Age agro-pastoral communities established thousands of permanent river-terrace hamlets across the forest-steppe from the Dnieper to the Yenisei. Archaeological registries catalogue 9.762 Bronze Age sites in Russia and 6.980 in Belarus and Ukraine. Mixed farming took root across the Ukrainian and Belarusian woodlands, while the Koban metallurgical culture constructed agricultural terraces in the central Caucasian valleys [43][44][45]. By 1000BC, total population reached 2,3M: 925k across the four Rus regions, 379,26k on the steppe and Crimea, 376k in the North Caucasus, 278k in the Russian forest core, and 337,64k across Siberia.

During the Scythian and Classical Greek eras (600BC to 300AD; Figure 4, Panel B), grain-exporting maritime routes connected Black Sea coastal poleis (Olbia, Chersonesos, Panticapaeum and Tanais) with fortified inland agricultural centres such as Belsk in the Vorskla basin [22][46]. Total numbers reached 3,28M by 500BC. At the turn of the era (1AD) and into the early Roman period (100AD), Sarmatian

pastoralists succeeded the Scythians on the open steppe, while Late Scythian farming in Crimea, Zarubintsy hamlets along the Dnieper, and Dyakovo hillforts in the Oka-Volga woodlands raised domain numbers to 3,63M in 1AD and 3,92M in 100AD [47][48].

Pre-mediaeval population peaked at 4,21M around 300AD under the Chernyakhov culture. Settlements concentrated heavily in the forest-steppe: Kiev-Pereyaslavl held 680k, Galicia-Volhynia 650k, Chernigov-Seversk 225,5k, and Polotsk-Turov 240k [32][49].

The Migration Period (375-550AD; Figure 4, Panel C) brought sharp contraction. Hunnic and Avar raids, coupled with volcanic cooling and plague in the sixth century, lowered population on the open steppe and Crimea by 27,3% (from 639,62k in 300AD to 465,03k in 500AD). The southern forest-steppe declined by 20,6% to 1,43M. By contrast, the northern mixed-forest zone held firm: population in the Russian core rose from 780,52k to 920k between 300AD and 500AD as Baltic, Slavic and Finno-Ugric communities sheltered within the riverine woodlands [50][51].

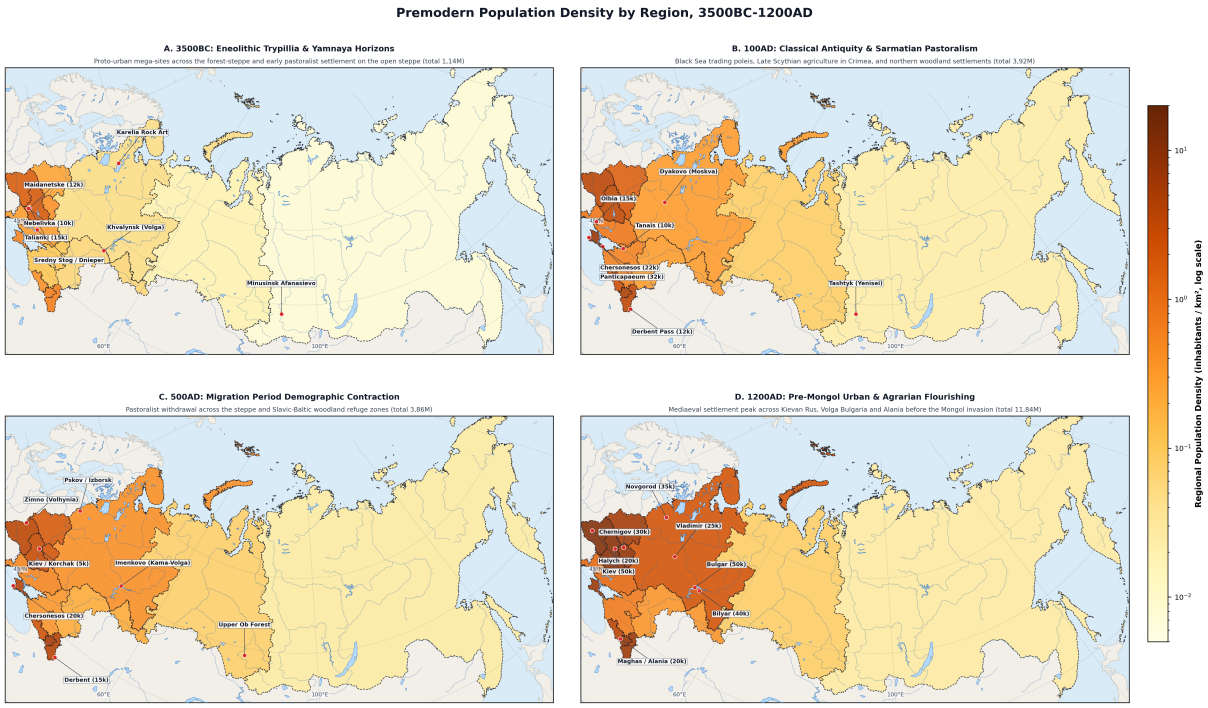


Figure 4. Pre-modern regional population density choropleths across the 14 constituent specialist divisions at 3500BC (A), 100AD (B), 500AD (C) and 1200AD (D). Points denote major excavated centres and towns.

Rise of Kievan Rus, Volga Bulgaria, Khazaria-Alania and the Pre-Mongol Apogee (600AD-1240AD)

Between 600AD and 1000AD, political stability under the Khazar Khaganate along the Lower Volga (Atil), Don (Sarkel) and Dagestan (Derbent) fostered the Saltovo-Mayaki agro-pastoral complex and Alanian urbanism in the North Caucasus [52][53]. North of the steppe, river trade routes linking the Baltic, Volga and Black Sea stimulated the rise of Kievan Rus and Volga Bulgaria. Total population grew from 4,7M in 700AD to 6,49M in 900AD and 7,74M in 1000AD, with 3,1M in the four Kievan Rus lands, 2,51M in the Russian forest core, 865k in the North Caucasus, and 743,63k across the steppe and Crimea (Figure 2, Panel C).

During the high mediaeval expansion (1000-1240AD; Figure 4, Panel D), the adoption of the heavy plough

on chernozem soils and the three-field rye rotation in northern woodlands sustained rapid agrarian growth [24][54]. By 1200AD, domain population reached a pre-modern peak of 11,84M (0,661 inhabitants/km^2).

The southern and western Rus lands held 4,88M people across 560k km^2 (8,71/km^2). Galicia-Volhynia reached 1,65M (anchored by Halych, Volodymyr and Peremyshl), Kiev-Pereyaslavl supported 1,45M (anchored by Kiev with 50k inhabitants), Polotsk-Turov reached 1,08M, and Chernigov-Seversk held 700k [25][55].

The north-eastern lands, Novgorod and Volga Bulgaria supported 4,67M across 2,19M km^2 (2,13/km^2). Forest clearance in the Vladimir-Suzdal Opolye, Ryazan, Smolensk and the Novgorod-Pskov territory accompanied urban development in Volga Bulgaria, where Bilyar and Bulgar commanded an agricultural hinterland of over 650k inhabitants [26][56].

The North Caucasus, Crimea and the Kipchak Steppe held an additional 1,82M people. The Christian kingdom of Alania (centred on Maghas), Highland Dagestan polities and Derbent supported 952k in the Caucasus, while Crimean trading cities and nomadic pasture groups accounted for 863,25k [57].

The Mongol Shock, Golden Horde Urbanism and the North-Eastern Shift (1240AD-1400AD)

The Mongol invasion of 1236-1242AD fundamentally altered regional settlement. Excavations across 74 fortified Rus towns demonstrate that 49 were destroyed during the conquest, of which 14 were never reoccupied and 15 were reduced to open rural villages [24]. Between 1200AD and 1300AD, total domain population contracted by 15% from 11,84M to 10,06M (Table 3), with marked regional disparities.

The southern forest-steppe suffered the heaviest losses, declining by 29,5%. Kiev-Pereyaslavl fell by 44,8% (from 1,45M to 800k, with Kiev dropping from 50k to under 8k), Chernigov-Seversk dropped by 40% (to 420k), and Galicia-Volhynia declined by 26,1% (to 1,22M). By contrast, forest-sheltered Polotsk-Turov fell by only 7,4%. In the North Caucasus, military campaigns and the destruction of Maghas lowered regional numbers from 952k to 750k [25][57].

Conversely, the Lower Volga and Crimea experienced an urban boom under the Golden Horde, expanding by 27,4% from 863,25k in 1200AD to 1,1M in 1300AD. Extensive artisan and caravan cities developed at Sarai Batu (~75k), Sarai Berke (~100k), and Ukek, while Caffa, Tana and Solkhat flourished in Crimea [27][58]. This urban network persisted until disrupted by the Black Death (1346-1353AD), civil conflicts, and Timur's campaigns in 1395AD [59][60].

In the northern Russian woodlands, agricultural settlement remained relatively stable. While Vladimir and Ryazan were sacked in 1237-1238AD and suffered plague outbreaks in 1352-1353AD, the spared north-western lands of Novgorod and Pskov together with monastic clearance in the Moscow, Tver and Kostroma basins kept the core at 4,33M in 1300AD, recovering to 4,85M by 1400AD [34].

Muscovite Consolidation, the Polish-Lithuanian Borderlands and the Conquest of Siberia (1400AD-1721AD)

From 1400AD onward, POLARIS follows the 38 regions of the Tsardom of Russia and its immediate borderlands across 18,12M km^2 (Tables 4 and 5, and Figure 5, Panel A). Between 1400AD and 1550AD, domain population rose from 11,36M (0,627/km^2) to 14,29M in 1500AD and 16,21M in 1550AD. In the Grand Duchy of Lithuania and the Polish Crown, the agrarian reform of 1557AD established uniform three-field holdings across Belarus and Volhynia, while the 1569AD Union of Lublin encouraged colonisation in Left-Bank and Right-Bank Ukraine, reaching 3,85M in Ukraine and 1,68M in Belarus by

1600AD [61][62]. Muscovite annexation of Novgorod, Kazan and Astrakhan unified the Volga trade route, lifting total population to 17,78M by 1600AD.

The seventeenth century witnessed significant demographic disruption followed by frontier expansion. The famine of 1601-1603AD, military raids and Polish-Swedish intervention during the Time of Troubles depopulated the Novgorod north-west and the Moscow centre, reducing total domain numbers from 17,78M in 1600AD to 16,8M in 1615AD [11][63].

After recovering to 20,7M in 1648AD, the Khmelnytsky Uprising, the Russo-Polish War (1654-1667AD), the Swedish Deluge and the 1654-1657AD plague devastated Belarus and Right-Bank Ukraine. The 1667AD hearth registers indicate that Belarus lost 45-50% of its pre-war households, pulling the domain total down to 18,6M [61][64].

From the 1630s onward, construction of the Belgorod, Simbirsk and Izium fortified lines shielded the Central Black-Earth region and Sloboda Ukraine from slave raids, allowing over 1,8M peasants and Cossacks to settle the fertile Voronezh-Kursk-Belgorod belt [65][66]. By the 1678AD Household Census (19,96M total; Figure 5, Panel A) and the First Revision in 1719-1721AD (22,83M in 1720AD), settlement had expanded substantially. East of the Urals, garrison forts established at Tobolsk, Tomsk, Yeniseysk, Yakutsk and Irkutsk anchored agricultural colonisation in the arable river basins, where Russian settlers (~310k by 1719AD) came to outnumber the indigenous tribute-paying population (~220k), bringing the Siberian total to 530k [17][67].

National Epoch & Scope	Start Year	Start Pop.	End Year	End Pop.	Urban Share
Tsardom of Russia (38 reg., 18,12M km^2)	1400AD	11,36M	1720AD	22,83M	3,8% to 4,8%
Early Russian Empire (40 reg., 18,5M km^2)	1721AD	23,89M	1815AD	51,17M	4,8% to 6,4%
Late Russian Empire (48 reg., 18,61M km^2)	1815AD	57,87M	1914AD	167,95M	6,5% to 15,8%
First World War & Civil War (48 reg., 18,61M km^2)	1914AD	167,95M	1926AD	167,78M	15,8% to 17,9%
Soviet Union (Lenin & Stalin) (45 reg., 18,18M km^2)	1926AD	145,07M	1953AD	165,35M	17,9% to 43,4%
Soviet Union (Post-Stalin) (41 reg., 18,18M km^2)	1953AD	165,35M	1991AD	224,55M	43,4% to 66,8%
Post-Soviet Space (35 reg., 17,97M km^2)	1991AD	212,26M	2021AD	198,08M	67,1% to 73,9%
War in Ukraine & Three States (123 div., 17,96M km^2)	2021AD	198,08M	2026AD	188,56M	73,9% to 74,8%

Table 4. National epoch population trajectories and urban shares across Eastern Europe and Northern Asia from 1400AD to 2026AD.

National Epoch Module	Event / Shock Horizon	Benchmark Pop. / Trough	Territorial Splices & Demographic Shocks
Tsardom of Russia	1601-1603AD	16,8M	Oprichnina, 1601-1603 famine and Time of Troubles (-980k trough)
Tsardom of Russia	1654-1667AD	18,6M	Khmelnytsky Uprising, Swedish Deluge and 1654-1657 plague (-2,1M)

Early Russian Empire	1762AD	32,34M	1st to 3rd Revisions; Ural metallurgical boom and Sloboda Ukraine growth
Early Russian Empire	1795-1812AD	49,49M	Polish Partitions (+7,3M), Crimea (1783), Finland (1809), Bessarabia (1812)
Late Russian Empire	1815AD	57,87M	+6,7M Congress Poland boundary splice; 1861 peasant emancipation
Late Russian Empire	1897-1914AD	123,89M	Donbas-Kryvyi Rih coal and steel boom; 3,8M Trans-Siberian agrarian migration
First World War & Civil War	1922AD	148,66M	First World War casualties, Civil War, typhus, and 1921-1922 Volga famine
Soviet Union (Lenin & Stalin)	1933AD	149,93M	-22,71M Finland and Poland exit splice; 1932-1933 Holodomor famine
Soviet Union (Lenin & Stalin)	1945AD	142,62M	Second World War invasion trough (-29,96M net loss); 1946-1947 drought famine
Soviet Union (Post-Stalin)	1970AD	197,91M	Post-war demographic recovery, Virgin Lands campaign, and West Siberian oil cities
Post-Soviet Space	1991-2005AD	203,24M	-12,29M Baltic and Moldovan exit splice; post-Soviet male mortality crisis
War in Ukraine	2023-2026AD	188,56M	Full-scale Russian invasion (6,3M refugees abroad, 6,2M in occupied Ukraine)

Table 5. Major demographic crises, famine troughs and territorial boundary splices across the national epoch series from 1400AD to 2026AD.

Imperial Expansion, Revision Censuses, Novorossiya Colonisation and Industrialisation (1721AD-1914AD)

The proclamation of the Russian Empire in 1721AD and the institution of the poll tax initiated a series of ten revisions (*revizii*, 1719-1858AD) [11][16]. Across the early imperial domain (40 regions, 18,5M km^2), population more than doubled from 23,89M in 1721AD (First Revision) to 27,49M in 1744AD, 32,34M in 1762AD, 38,42M in 1782AD, 43,7M in 1795AD, 49,49M in 1811AD, and 51,17M in 1815AD (Table 4 and Figure 3, Panel B).

Territorial annexations and agrarian colonisation reinforced this growth. The three partitions of the Polish-Lithuanian Commonwealth (1772, 1793, 1795AD) incorporated western borderlands (~7,3M inhabitants in 1795AD) into the imperial fiscal system. The subsequent acquisitions of Finland (1809AD), Bessarabia (1812AD) and Congress Poland (1815AD) added further populations, with the 1815AD Polish entry adding 6,7M inhabitants to the 48-region series [12][68].

Concurrently, military victories over the Ottoman Empire and the annexation of Crimea (1783AD) opened the southern steppe to agricultural settlement. Immigrant colonists and peasant migrants founded

Odessa (1794AD, reaching 404k in 1897AD and 631k in 1914AD) and settled Kherson and Ekaterinoslav [69][70]. In the North Caucasus, the protracted Caucasian War and the 1864AD Circassian expulsion reduced indigenous populations, followed by Cossack and peasant resettlement that brought regional population to 3,72M by 1897AD and 5,45M by 1914AD [71][72].

Following the emancipation of the serfs in 1861AD (83,92M total at the Tenth Revision in 1858AD), sustained high fertility generated severe rural land pressure across the Central Black-Earth and Middle Volga provinces [73]. The rapid development of the Donbas coal and metallurgy basin and the construction of the Trans-Siberian Railway stimulated industrial urbanisation and eastern migration, moving over 3,8M peasant settlers across the Urals [74][75]. At the 1897 All-Russian Census [18], domain population reached 123,89M (13,4% urban), climbing to 167,95M (15,8% urban; Figure 5, Panel B) on the eve of the First World War in 1914AD.

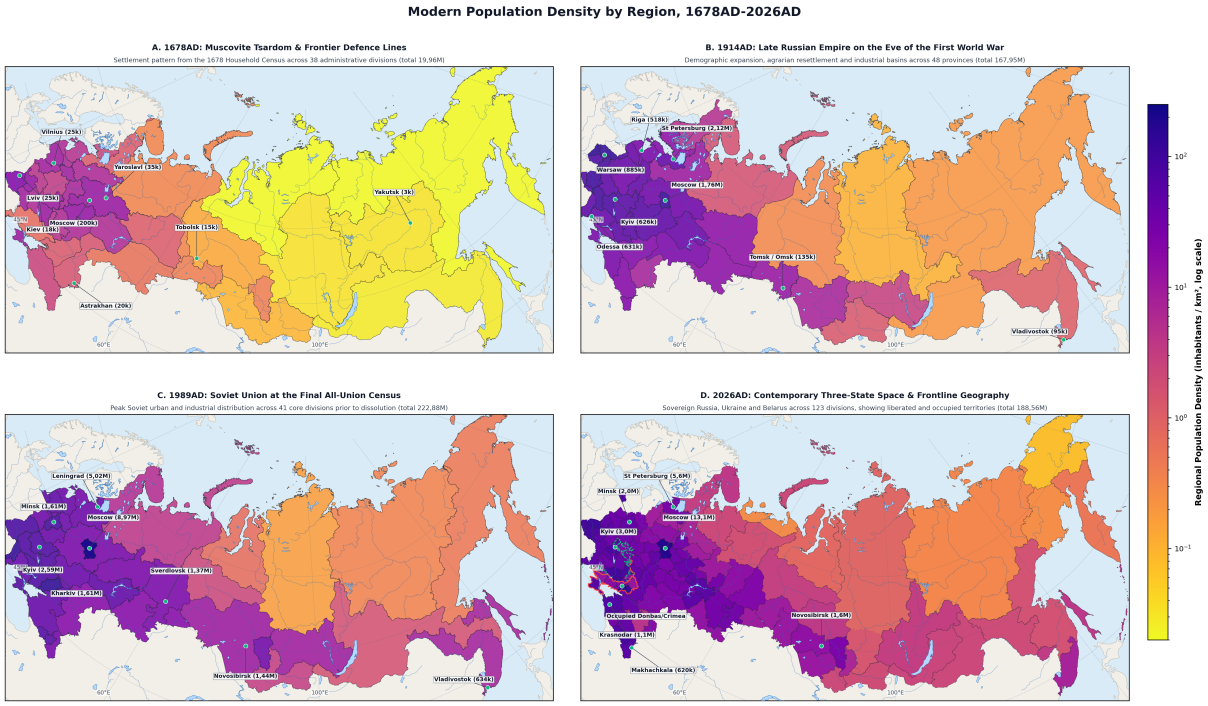


Figure 5. Modern regional population density choropleths and urban centres at 1678AD (A), 1914AD (B), 1989AD (C) and 2026AD (D). Panel D marks liberated and occupied Ukrainian territories from DeepStateMAP.

Demographic Catastrophe and Soviet Modernisation: War, Famine and Urbanisation (1914AD-1991AD)

Between 1914AD and 1947AD, Eastern Europe and Northern Asia endured three devastating demographic shocks separated by rapid, state-directed industrialisation (Tables 4 and 5, and Figure 3, Panels B and D).

During the First World War and Civil War (1914-1926AD), military mobilisation, wartime mortality, epidemic typhus, emigration and the 1921-1922AD famine across the Volga-Ural and Ukrainian agricultural zones reduced total population from 167,95M in 1914AD to a trough of 148,66M in 1922AD [19][76][77]. Major cities experienced substantial de-urbanisation as food supply collapsed: Petrograd lost two-thirds of its inhabitants, and Moscow halved. Under the New Economic Policy, high rural birth rates restored the 48-region total to 167,78M by December 1926AD.

The interwar Soviet baseline (1926-1941AD) begins at 145,07M (17,9% urban) following the departure of

independent Finland and interwar Poland (-22,71M splice in 1926AD) [78]. Forced collectivisation, peasant deportations and state grain requisitions precipitated the 1932-1933AD famine, which caused 5,7M to 7M excess deaths across the Soviet Union, including 3,5M to 3,9M in Ukraine (Holodomor) and 1,8M to 2,5M in southern Russia, the Urals and Kazakhstan [79][80]. Total population dipped from 153,85M in 1930AD to 149,93M in 1933AD, recovering to 166,2M in 1939AD and 172,58M (33,2% urban) in mid-1941AD after the Soviet annexation of western Ukraine, western Belarus, the Baltic republics and Bessarabia [20].

The Second World War (1941-1945AD) inflicted the deepest demographic contraction in the historical record. Total population fell by 29,96M (-17,4%) from 172,58M in 1941AD to 142,62M in 1945AD, driven by military casualties, civilian deaths under occupation, the Holocaust, siege starvation in Leningrad, and wartime deportations [19][81][82]. Belarus lost over a quarter of its pre-war population, falling from 9,05M in 1940AD to 6,28M in 1945AD, while Ukraine contracted from 41,3M to 31,8M before absorbing further losses in the 1946-1947AD drought famine [83][84]. Wartime industrial evacuation permanently shifted manufacturing capacity east of the Urals.

In the post-Stalin era (1953-1991AD), public health improvements, housing construction and rural-to-urban migration accelerated modernisation across the 41-region Soviet domain. Urban population crossed 50% in the early 1960s, reaching 56,6% in 1970AD (197,91M total) and 66,4% in 1989AD (222,88M; Figure 5, Panel C) [85][86]. While rural populations in central European Russia experienced persistent decline, substantial resource investments expanded urban centres in the West Siberian oil and gas basins and along the Baikal-Amur Mainline. Domain population peaked at 224,55M in 1991AD.

Post-Soviet Demographic Transition and the Russian Invasion of Ukraine (1991AD-2026AD)

The dissolution of the Soviet Union in December 1991AD established a new spatial baseline. The exit of the Baltic states and Moldova accounts for a 12,29M administrative splice, reducing the core three-state population (Russia, Ukraine and Belarus) from 224,55M to 212,26M in 1991AD (Table 6).

Between 1991AD and 2021AD, the region experienced a prolonged peacetime demographic contraction, declining by 14,18M (-6,7%) to 198,08M in 2021AD. Economic dislocation in the 1990s drove adult male mortality sharply upward and pushed fertility rates down to 1,1-1,25 children per woman [87][88]. Out-migration depleted northern and Far Eastern resource towns, while population within Russia concentrated increasingly into Moscow, St Petersburg, Krasnodar Krai and the North Caucasus [89][90]. In Ukraine, the 2014AD Russian annexation of Crimea and the Donbas conflict displaced over 1,5M people and reduced resident population within 1991 sovereign borders from 51,5M in 1991AD to 43,87M in 2021AD [14].

The full-scale Russian invasion launched on 24 February 2022 generated the largest refugee and displacement crisis in Europe since the Second World War [14][15]. Frontline mapping from DeepStateMAP shows that Ukrainian counter-offensives in late 2022AD liberated approximately 42.000 km^2 across Northern Ukraine, Kharkiv and Right-Bank Kherson, leaving roughly 109.000 km^2 (~18% of Ukrainian sovereign territory) under Russian occupation in 2024-2026AD (Figure 5, Panel D).

Between 2021AD and 2026AD, total population across the three countries declined from 198,08M to 188,56M (-4,8%). Ukraine within sovereign 1991 borders contracted from 43,87M to 37,66M (-14,2%), comprising approximately 31,44M in government-controlled territory and 6,22M in Russian-occupied regions. This drop reflects 6M to 6,5M external refugees, 3,7M internal displacements, wartime casualties, and a fertility decline to 0,8-0,9 children per woman [14][15]. The Russian Federation within 1991 borders declined from 145,06M to 141,79M (or 144,19M including Crimea), driven by elevated natural decrease, pandemic excess mortality, war casualties, and outward migration [91][92]. In Belarus, population edged down from 9,15M to 9,11M [93].

Sub-Polity/Territorial Zone	1897AD	1926AD	1959AD	1989AD	2021AD	2026AD
Russian Federation (1991 borders excl. Crimea)	67,47M	92,74M	117,53M	147,02M	145,06M	141,79M
Ukraine (1991 sovereign borders incl. Crimea)	28,45M	36,82M	41,87M	51,45M	43,87M	37,66M
- Kyiv-controlled Ukraine (incl. 2022 liberated zones)	24,1M	31,2M	34,2M	41,8M	37,6M	31,44M
- Russian-occupied Ukraine (Crimea + Donbas/South)	4,35M	5,62M	7,67M	9,65M	6,27M	6,22M
Republic of Belarus (1991 sovereign borders)	6,67M	7,15M	8,06M	10,15M	9,15M	9,11M
Core 3-State Subtotal (RU + UA + BY, 17,96M km^2)	102,59M	136,71M	167,46M	208,62M	198,08M	188,56M
Baltic States (Estonia, Latvia, Lithuania) & Moldova	7,28M	8,36M	9,94M	14,26M (exit 91)	-	-
Grand Duchy of Finland & Congress Poland (1815-1917 scope)	14,02M	22,71M (exit)	-	-	-	-

Table 6. Harmonised modern sub-polity population breakdown across Russia, Ukraine, Belarus and historical borderlands from 1897AD to 2026AD.

Limitations

Historical demographic reconstructions carry unavoidable uncertainties that vary with the nature of surviving sources [1][3]. Four specific caveats apply to the regional figures presented here.

First, pre-documentary population totals prior to 1500AD rely on archaeological site densities, house-floor counts, walled town dimensions and carrying-capacity models. While the kriging mosaic constrains relative spatial distributions across raions, absolute regional population totals carry uncertainty bands of ±35% to 50% during the prehistoric periods (10000-1000BC) and ±20% to 30% during Classical Antiquity and the Middle Ages (Figure 3, Panel A).

Second, archaeological survey coverage varies considerably across administrative regions. In the combined dataset, 34.365 monuments (31,1%) are registered without specific chronological subdivisions, and 2.393 records lack precise raion coordinates. While Bayes shrinkage mitigates regional recording bias, alluvial sedimentation in southern river deltas and thermokarst erosion in northern Siberia likely lead to under-representation of early Holocene sites.

Third, converting early modern fiscal cadastres and hearth registers into total population requires household multipliers that fluctuated over time. Peasant households frequently merged into extended family compounds to minimise household-based tax liability [11][94]. In addition, the earliest revisions

under-recorded female populations in certain borderlands and missed fugitive serfs along frontier zones, requiring retroactive adjustments.

Fourth, contemporary demographic statistics for 2022-2026AD operate under wartime reporting restrictions. Current estimates for Ukraine rely on a synthesis of international displacement registers, border-crossing data, mobile subscriber counts and municipal reports [14][15]. Figures for Russian-occupied territories (~6,22M in 2026AD) remain subject to ±10% to 15% uncertainty until official post-conflict enumerations become possible.

Conclusion

The population history of Eastern Europe and Northern Asia over the past twelve millennia is the story of human expansion into vast ecological spaces, repeatedly redirected by climatic stress and political catastrophe.

At the end of the last Ice Age (10000BC), fewer than 90.000 hunter-gatherers occupied this enormous landmass (0,005/km^2). The adoption of farming and pastoralism accelerated growth during the Eneolithic, producing the remarkable proto-urban mega-sites of the Cucuteni-Trypillia culture in the right-bank forest-steppe (1,14M total in 3000BC) alongside the mobile pastoralism of the Yamnaya horizon.

In Classical and Late Roman times, arable farming, trade with Black Sea Greek poleis, and the Chernyakhov culture lifted domain numbers to an initial peak of 4,21M in 300AD. The nomadic incursions, sixth-century climatic cooling and plague of the Migration Period pulled population back to 3,86M by 500AD.

High mediaeval recovery, powered by the heavy plough on the southern chernozem margin and the three-field rotation in the northern woodlands, raised the population of Kievan Rus, Volga Bulgaria and Alania to 11,84M by 1200AD. The Mongol conquests of 1236-1242AD broke this trajectory, reducing population by 15% (to 10,06M in 1300AD) and permanently shifting the settlement balance: south-western Rus principalities contracted sharply, while the northern forest core held firm and the Lower Volga urbanised under the Golden Horde.

From 1400AD onward, fortified frontier lines and imperial expansion opened the Central Black-Earth region, the southern steppe, the North Caucasus, and the Siberian river basins to mass agricultural colonisation. Total numbers rose from 11,36M in 1400AD to 22,83M in 1720AD, 51,17M in 1815AD, and 167,95M on the eve of the First World War in 1914AD.

The twentieth century brought profound demographic upheaval: combat losses, the 1921-1922 and 1932-1933 famines, the devastation of the Second World War, and rapid forced industrialisation. Domain numbers peaked at 224,55M in 1991AD. The subsequent post-Soviet transition, marked by sharp fertility decline and heightened adult mortality, followed by the 2022-2026AD war in Ukraine, reduced the combined population of Russia, Ukraine and Belarus to 188,56M by 2026AD.

Sources

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